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DCME C-23 SYLLUBU
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Certainly ! Below is a sample question paper based on the syllabus and model papers provided in the files . The questions are designed to cover the key topics and learning outcomes mentioned in the syllabus .

STATE BOARD OF TECHNICAL EDUCATION, A.P.

ENGINEERING MATHEMATICS -I (C-102)

TIME : 3 HOURS

MAX . MARKS : 80M

PART-A

Answer All questions . Each question carries THREE marks .

10 x 3 = 30M

1. Algebra :

If $A = \{-1, 0, 1\}$ and $f : A \rightarrow B$ is defined by $f(x) = x^2 - x + 1$, find the range of f . (C01)

2. Partial Fractions :

Resolve the function $\frac{1}{(x+1)(x-2)}$ into partial fractions . (C01)

3. Matrices :

If $A = \begin{bmatrix} 3 & 9 & 0 \\ 1 & 8 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 0 & 2 \\ 7 & 1 & 4 \end{bmatrix}$, find $(A + B)^T$. (C01)

4. Trigonometry :

Prove that $\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$. (C02)

5. Trigonometry :

Find the value of $\frac{\cos 36^\circ + \sin 36^\circ}{\cos 36^\circ - \sin 36^\circ}$. (C02)

6. Complex Numbers :

Find the modulus of the complex number $3 + 2i$. (C02)

7. Coordinate Geometry :

Find the point of intersection of the lines $x + y + 1 = 0$ and $2x - y + 5 = 0$

Find the point of intersection of the lines $x + y + 1 = 0$ and $2x - y + 3 = 0$.
(C03)

8. **Limits:**

Evaluate $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$. (C04)

9. **Differentiation :**

Find $\frac{dy}{dx}$, if $y = x^3 + 5x$. (C04)

10. **Differentiation :**

Find the derivative of $3 \sin x + \log x + 2 \tan^{-1} x + 8e^{-x}$ with respect to x .
(C04)

PART-B

Answer any FIVE questions. Each question carries TEN marks.

5 x 10 = 50M

11. **Algebra :**

Solve the system of linear equations $x - y + 3z = 5$, $4x + 2y - z = 0$, and $-x + 3y + z = 5$ using Cramer's rule. (C01)

12. **Trigonometry :**

A) Show that $\cos 40^\circ + \cos 80^\circ + \cos 160^\circ = 0$. (C02)

B) Prove that $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$. (C02)

13. **Trigonometry :**

A) Solve $2 \cos^2 \theta - 3 \cos \theta + 1 = 0$. (C02)

B) If $a = 5$, $b = 7$, and $c = 30^\circ$, find the area of triangle ABC . (C02)

14. **Coordinate Geometry :**

A) Find the equation of the circle passing through the points $(0, 0)$, $(2, 0)$, and $(0, 3)$. (C03)

B) Find the vertex, focus, directrix, and latus rectum of the parabola $y^2 = 16x$.
(C03)

15. **Differentiation :**

A) If $y = x^{\sin x}$, find $\frac{dy}{dx}$. (C04)

B) If $y = \tan^{-1} x$, prove that $(1 + x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} = 0$. (C04)

16. **Applications of Derivatives :**

A) Find the equation of the tangent to the curve $y = x^3 - 2x^2 + 4$ at the point $(2, 4)$. (C05)

B) If $s(t) = t^2 + 2t + 3$ is the displacement of a particle find its velocity and

where $s(t) = t^3 + 2t^2 + 3t + 5$ is the displacement of a particle, and its velocity and acceleration at $t = 3$ sec. (C05)

17. Applications of Derivatives :

A) Find the maximum or minimum value of $f(x) = 3 + 10x - 5x^2$. (C05)

B) If an error of 0.02 cm is made in the side of a square, find the approximate error in the area and perimeter of the square. (C05)

18. Applications of Derivatives :

A) Find the rate of increase in the area of a circular plate when the radius is 10 cm and the radius is increasing at 0.7 cm/sec. (C05)

B) Find the percentage error in the calculated area of a square if an error of 0.02 cm is made in its side. (C05)

END OF QUESTION PAPER

This question paper is designed to cover the key topics from the syllabus, including **Algebra, Trigonometry, Coordinate Geometry, Differential Calculus**, and **Applications of Derivatives**. The questions are aligned with the learning outcomes and course objectives mentioned in the syllabus.